

Quiz: Basic arithmetic

This quiz has 5 questions. The passing score is 4.

[Back](#)[Next](#)

Started on	Tuesday, January 14, 2025, 4:14 PM
State	Finished
Completed on	Tuesday, January 14, 2025, 4:23 PM
Time taken	9 mins 1 sec
Grade	5.00 out of 5.00 (100%)

Question **1**

Correct

1.00 points out of 1.00

Given two fullwords, 000ABCDE and 00054321, adding them results in the fullword:

Select one:

- ☐ A. FFFFFFFF
- ☐ B. The result can't be represented in a fullword
- ☐ C. ABCDE000
- ☒ D. 000FFFFF ✓
- ☐ E. FFFF000

Correct!

```
000ABCDE
+ 00054321
-----
000FFFFF
```

The correct answer is: 000FFFFF

Question **2**

Correct

1.00 points out of 1.00

Back

Next

(Assuming the numbers are fullword integers): **00000010 - 000F0010 = FFF10000**

Select one:

☒ True ✓☐ False

Correct!

000F0010 (decimal **983,056**) is bigger than **00000010** (decimal **16**), so we'll calculate **000F0010 - 00000010** and change the sign of the result.

```

000F0010
- 00000010
-----
000F0000 decimal 983,040

```

To change the sign of **000F0000**, we invert the bits and add 1:

```

000F0000
FFF0FFFF (bits inverted)
    + 1
    ----
FFF10000 result of 00000010 - 000F0010 (decimal
-983,040)

```

The correct answer is 'True'.

Question **3**

Correct

1.00 points out of 1.00

[Back](#)[Next](#)

A halfword containing hexadecimal 5AAA is, in decimal:

Select one:

- ☐ A. 5,888
- ☐ B. -200,300
- ☐ C. 20,000
- ☒ D. 23,210 ✓
- ☐ E. 5,101,010

Correct!

Hexadecimal 5AAA to decimal:

$$5 \times 16^3 + 10 \times 16^2 + 10 \times 16^1 + 10 \times 16^0 =$$

$$5 \times 4096 + 10 \times 256 + 10 \times 16 + 10 \times 1 =$$

$$20480 + 2560 + 160 + 10 = \mathbf{23,210}$$

The correct answer is: 23,210

Question **4**

Correct

1.00 points out of 1.00

Back

Next

Calculate the hexadecimal addition $70A0 + 0A0A$. The result can be represented as (pick all that apply):

Select one or more:

- ☐ A. Halfword FAAA
- ☐ B. Fullword FFFF7AAA
- ☒ C. Doubleword 0000000000007AAA ✓
- ☒ D. Halfword 7AAA ✓
- ☒ E. Fullword 00007AAA ✓

Correct!

70A0
+ 0A0A

7AAA, or binary **0111 1010 1010 1010** (sign bit 0)

This number is already **halfword 7AAA**.

To represent it in a **fullword**, we propagate the sign (0) to the left: **00007AAA**.

Same for **doubleword**: **0000000000007AAA**.

The correct answers are: Halfword 7AAA, Doubleword 0000000000007AAA, Fullword 00007AAA

Question **5**

Correct

1.00 points out of 1.00

[Back](#)[Next](#)

The decimal number 8,000 can be represented as (pick the two correct answers)

Select one or more:

- ☒ A. Halfword **1F40** ✓
- ☐ B. Halfword E0C0
- ☒ C. Fullword **00001F40** ✓
- ☐ D. Fullword FFFFE0C0
- ☐ E. Fullword 0000E0C0

Correct!

Converting decimal 8,000 to hexadecimal:

8000 / 16 = 500, with remainder **0**
500 / 16 = 31, with remainder **4**
31 / 16 = 1, with remainder 15 (hex **F**)
1 / 16 = 0, with remainder **1**

Decimal 8,000 is hexadecimal **1F40**. This is:

- Halfword **1F40**
- Fullword **00001F40**

The correct answers are: Halfword 1F40, Fullword 00001F40